



Python Programming Language Syllabus



Introduction to Python

- History and features of Python
- Installing Python and IDEs (IDLE, VS Code, Jupyter)
- Writing and executing Python scripts
- Python syntax, comments, Indentation
- Variables and data types
- Type casting and input/output functions



Control Flow Statements

- if, if-else, if-elif-else
- while loop, for loop
- break, continue, pass statements
- Loop control and nested loops



Data Structures

- Strings (slicing, methods, formatting)
- Lists (indexing, slicing, methods)
- Tuples (immutable sequences)
- Sets (unique elements, operations)
- Dictionaries (key-value pairs)



Functions

- Defining and calling functions
- Function arguments: default, keyword, variable-length
- Return statement
- Recursion
- Lambda functions, map(), filter(), reduce()



Introduction to Libraries & Projects

- Basic intro to
 - NumPy (arrays, numerical operations)
 - Pandas (data manipulation)
 - Matplotlib (plotting)
- Mini projects (Calculator, To-Do List, Quiz App, etc.)



Functions

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- Function arguments: default, keyword, variable-length
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Object-Oriented Programming

- Classes and objects
- Constructor (.__init__) and destructor
- Inheritance and types
- Method overriding
- Encapsulation • data hiding



File Handling

- Types of errors
- try, except, else, finally blocks
- Built-in exceptions
- User-defined exceptions



Modules and Packages

- Importing built-in modules (math, random, datetime)
- Creating and using user-defined modules
- Installing external packages using pip
- Using `__name__ == '__main__'`

Recommended Study Materials Books

- "Python Crash Course" by Eric Matthes
- "Automate the Boring Stuff" by Al Sweigart
- "Learning Python" by Mark Lutt